NEURALPATH CHEATSHEET

Module 4: Computer Vision

Explore convolutions, residual networks, augmentation, transfer learning, Vision Transformers (

Core Formulas, Mental Models & Engineering Rules

- Conv2D output size: floor((H + 2p - k) / s) + 1.
- ResNet skip connections y = F(x) + x eliminate vanishing gradients in deep networks.
- Transfer Learning: freeze early edge/texture layers; fine-tune top blocks with lower LR.
- ViT splits images into fixed patches + positional embeddings and applies self-attention.
- Detection/Segmentation: YOLO/DETR for bounding boxes; U-Net and SAM for dense pixel masks.

Estimated study time: 4 hours · Badge: Visual Perception Engineer